The ripple effects of disability insurance incentives: labor supply, benefit substitution, and spousal adjustment*

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Abstract

We examine whether financial incentives embedded in the Dutch disability insurance (DI) scheme induce partially disabled individuals to activate their residual work capacity. Exploiting exogenous variation in the timing of financial incentives, we document anticipatory responses and a sustained 5 percentage point (pp) increase in employment. These incentives simultaneously lead to increased claims from alternative social support programs. Furthermore, we document an added worker effect of up to 3 pp through spousal labor supply responses. The impact of financial incentives is stronger when the incentives are larger. Consistent with an added worker effect, spousal labor-supply responses are strongest in households facing larger income losses. Moreover, the effectiveness of financial incentives depends on labor market conditions, with responses being substantially larger when labor demand is high. We find no evidence that the labor-supply responses induced by financial incentives come at the expense of beneficiaries' health.

1 Introduction

In 2006, the Netherlands reformed its DI system, leading to a strong decrease in annual inflow into DI (Van Sonsbeek and Gradus, 2013; Kantarcı et al., 2023). However, inflow into the scheme for partially and temporarily fully disabled individuals increased from 17,300 in 2006 to 55,800 in 2024. With this increase, the stock of partially disabled individuals claiming benefits has reached 273,000 in 2024. There are several reasons for this growth, most notably the rising number of older workers and those without permanent contracts, groups with elevated DI risk. The population of partially disabled individuals is particularly relevant for policy analysis, as

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they retain residual work capacity and may be especially susceptible to moral hazard, given the difficulty of reassessing work capacity at scale. Targeted interventions could therefore be most effective if focused on this group.

When designing interventions to reduce the size of a DI scheme, two main policy approaches are available: limiting inflow into the program and reducing benefit use among beneficiaries. Inflow can be limited through stricter screening during the sickness insurance period or stricter eligibility criteria for DI, so that fewer insured workers qualify for benefits. Benefit use among beneficiaries can be reduced through stronger reintegration efforts. Financial incentives are another policy tool commonly used to reduce both inflow and benefit use.

A substantial body of literature has developed analyzing the causal effects of financial incentives. This work can be divided into two strands. The first strand studies the impact of financial incentives on inflow into DI. To limit inflow, schemes often reduce benefit generosity or tighten eligibility to discourage applications. Studies conducted in Austria, Canada, and the United States find that reducing benefit generosity reduces benefit claims and awards and increases labor participation, but the sizes of the estimated effects vary considerably (Bound and Burkhauser, 1999; Gruber, 2000; Mullen and Staubli, 2016; Favre et al., 2021). Campolieti (2004), however, finds no effect on DI claims or labor participation among older workers.

The second strand studies the impact of financial incentives on labor participation and benefit use of people in DI, including both increased work while on benefits and transitions out of DI through work. Many DI schemes allow earnings up to a certain threshold and reduce benefits if earnings exceed the threshold. Therefore, a large number of studies investigate the effects of changing the earnings threshold or the effects of reducing benefits at different rates when earnings exceed the threshold. With respect to changing the earnings threshold, in Canada, Campolieti and Riddell (2012) study the effect of changing the amount of wages that DI beneficiaries are allowed to earn without losing DI benefits. They find a substantial labor participation increase of 5 to 6 pp for men and 8 to 10 pp for women due to the introduction of the earnings allowance. Vall Castelló (2017) analyze a reform in Spain on income tax exemption for partially disabled beneficiaries who did not work. The reform eliminated the tax exemption for those aged 55 or younger but kept it for older workers. The reform increased labor participation of the younger group by 6.5 pp. In Austria, if earnings exceed a threshold amount, DI benefits are reduced by up to 50%. Ruh and Staubli (2019) find substantial and sharp bunching at the earnings threshold, and that the average DI recipient with earnings just below the threshold would increase earnings by 45% if the threshold was eliminated. In Canada, Zaresani (2020) exploits a policy reform that changed the earnings threshold to increase labor participation. Zaresani also finds substantial bunching, which remained persistent after changing the earnings threshold, suggesting substantial adjustment costs for disabled workers.

With respect to reducing benefits when earnings exceed a threshold amount, in the US, Weathers and Hemmeter (2011) study a pilot project that replaced the complete loss of DI benefits due to earning above a threshold amount with a gradual reduction of DI benefits by 50% of the amount earned above an earnings disregard level. They find that the number of DI recipients with earnings above the threshold amount increases by 25% after the policy change. Kostøl and Mogstad (2014) analyze a 2005 Norwegian reform which introduced a program where DI benefits are reduced by 60% of the earnings accumulated above an earnings threshold. They find a substantial positive effect on labor participation.

Our study focuses on the second strand of the literature, analyzing the impact of financial incentives on labor participation and benefit use among the existing stock of DI recipients. In the Netherlands, when admitted to the DI scheme, partially disabled individuals start receiving a DI benefit that incorporates in it the unemployment insurance (UI) benefit. The benefit is reduced at a rate of 70% if benefit recipients earn wages. The duration of the benefit depends

on the work history and expires after 3 to up to 24 months from the first day of DI receipt. After this period individuals face two financial incentives to increase labor supply. First, the UI component of the benefit expires and DI recipients can no longer claim UI. Second, if their earnings exceed an individual-specific threshold amount, their benefit is increased so that an increase in earnings leads to an increase in the benefit. The financial incentives are larger for benefit recipients who earn higher wages before they fall sick.

Koning and van Sonsbeek (2017) evaluate the impact of these financial incentives between 2006 and 2013. They find that labor force participation increases by 1.4 and 2.6 pp in the short and long run, respectively. They find no effect on full work resumption. These responses are modest relative to both the international evidence and the substantial size of the underlying financial incentives.

We revisit these effects using a substantially extended sample period and extend the analysis of Koning and van Sonsbeek (2017) along several additional dimensions. We also employ a new identification strategy that exploits exogenous variation in the timing of benefit reductions across individuals within a dynamic event-study framework for staggered treatment adoption. Unlike conventional Two-Way Fixed Effects (TWFE) estimators, which can be biased when treatment effects are heterogeneous or dynamic, our approach is robust to these complications.

We document anticipatory responses of about 1 pp prior to benefit expiration, an immediate 3 pp increase upon transition, and a sustained effect of 5 pp over an 18-month post-treatment period. These effects are substantially larger than those reported by Koning and van Sonsbeek (2017). These large effects motivate a broader examination of the consequences of disability benefit reductions, extending beyond average labor supply responses.

First, the incentives are larger for beneficiaries with higher pre-sickness wages, and we observe corresponding heterogeneity in labor supply responses across the pre-sickness wage distribution. Individuals earning at most 1.5 times the minimum wage increase labor participation by about 4 pp, while those earning at least 2.5 times the minimum wage respond with an increase of about 7 pp. Since beneficiaries cannot manipulate the size of the financial incentives, this pattern reflects an income effect: the larger the incentive, the stronger the response. The results therefore suggest that tying financial incentives to pre-sickness earnings concentrates activation effects among higher-wage beneficiaries, implying that the employment gains generated by the policy are unevenly distributed across the earnings distribution.

Second, we show that unemployed beneficiaries respond much more strongly to financial incentives than those with permanent or temporary employment contracts. Burkhauser et al. (2014) recognized that even individuals with severe impairments retain some capacity to work. Our finding extends this insight by showing that unemployed beneficiaries, despite being at greater risk of long-term labor market detachment, exhibit particularly strong labor-supply responses to financial incentives.

Third, we show that financial incentives are substantially more effective during periods of strong labor demand. This finding help explain why Koning and van Sonsbeek (2017) found only modest responses to these incentives. Our extended observation period covers 2006 to 2020 and includes years with substantially stronger labor market conditions following the financial crises of 2007-2010. The explanation is plausible because disabled individuals face limited job opportunities and are particularly vulnerable to deteriorating labor market conditions. More broadly, our findings suggest that the effectiveness of reintegration policies depends not only on their design but also on the prevailing macroeconomic environment. As a result, evaluations of such policies should explicitly account for business cycle conditions.

Fourth, an important concern is that stronger financial incentives may affect health. Using administrative records on prescribed medicines as a proxy for health status, we find no evidence that the transition to the less generous benefit regime leads to changes in medicine use. This

finding suggests that the labor-supply responses induced by financial incentives are unlikely to be driven by contemporaneous changes in health and alleviates concerns that increased work effort comes at the expense of beneficiaries' health.

Fifth, several studies have shown that major reforms limiting inflow into DI, typically through stricter eligibility criteria, lead sick individuals to substitute DI benefits with either earnings or support from alternative social programs (Karlström et al., 2008; Staubli, 2011; Kantarcı et al., 2023). Borghans et al. (2014) document similar substitution behavior following the reassessment of disability grades in the Dutch DI scheme in 1993. We extend this literature by showing that such responses do not require either stricter eligibility criteria or reassessments of existing recipients. We show that similar responses arise even when eligibility and disability assessments remain unchanged. Changes in benefit generosity alone are sufficient to induce both labor supply responses and substitution toward alternative social programs.

Finally, the literature on the added worker effect suggests that spouses, typically wives, may increase their labor market participation in response to a reduction in household income caused by their partner's unemployment. Bernasconi et al. (2024) examine this mechanism in the context of DI and document spousal labor supply responses following the tightening of DI eligibility criteria under the 2006 Dutch DI reform. We extend the literature on the added worker effect by providing strong evidence of spousal labor supply responses to benefit reductions among partially disabled individuals. Studies of unemployment shocks by Cullen and Gruber (2000) and Bentolila and Ichino (2008) argue that spousal labor supply responses are muted because affected workers are insured through alternative forms of social protection. Our findings instead suggest that the persistence of the income shock plays a central role in generating the added worker effect. Because disabled individuals have limited scope to restore lost income through their own labor supply, spouses increase labor market participation to help smooth household consumption, even when alternative forms of social support are available.

The remainder of this paper is organized as follows. Section 2 outlines the Dutch DI scheme for partially disabled individuals and the associated financial incentives to encourage work resumption. Section 3 describes the data. Section 4 presents descriptive statistics and preliminary evidence on the impact of financial incentives. Section 5 details the identification strategy. Sections 6 through 7 report the estimation results. Section 8 concludes.

2 The Dutch DI scheme and the financial incentives for work resumption

The Work and Income According to Labor Capacity Act (WIA) came into effect on 1 January 2006 for people who fell ill from 1 January 2004 onwards. In the WIA, individuals who lose any part of their earning capacity due to a health impairment are entitled to the sickness benefit from their employer for a period of two years. When the sickness benefit expires, they can apply for a DI benefit. If the wage loss is more than 80% and expected to be lasting, the individual is admitted to the Income Provision Scheme for Fully Disabled People (IVA). We do not study IVA recipients since their remaining earning capacity is very limited and odds of work resumption are very small. If the wage loss is more than 35% but less than 80%, or if the wage loss is more than 80% but there is a possibility of recovery, the individual is admitted to the Return to Work Scheme for Partially Disabled People (WGA).

The WGA consists of two stages. In the first stage the individual is entitled to the "wage-related benefit". The benefit is the sum of two components. The first component is a fixed amount (as long as the disability grade is not reassessed) and equal to

$$0.7 \times \text{pre–sickness wage} \times \text{disability grade}.$$

The second component is variable depending on how much the remaining work capacity is utilized and equal to

$$0.7 \times \text{pre–sickness wage} \times (1 - \text{disability grade}) \times (1 - \text{utilization rate}).$$

Utilization rate takes a value between 0 and 1, so that the last term represents the remaining earning capacity that is not utilized.[1] In fact, this component is the UI benefit embedded in the wage-related benefit that compensates individuals who do not utilize their remaining earning capacity. Duration of the wage-related benefit, for this reason, is equal to the duration of the UI benefit. It depends on the number of contribution years to social insurance, with each year of contributions granting half a month of benefit duration. The maximum duration is currently 24 months.[2] UI benefit entitlement (regardless of disability) is exhausted as the wage-related benefit is exhausted.

When the wage-related benefit expires, individuals become entitled to one of two types of second-stage benefits depending on whether they utilize at least 50% of their remaining earning capacity. If they utilize at least 50% of their remaining earning capacity, they become entitled to the "wage-supplement benefit" which is equal to

$$0.7 \times \text{pre–sickness wage} \times \text{disability grade}.$$

If they utilize less than 50% of their remaining earning capacity, they become entitled to the "follow-up benefit" which is equal to

$$0.7 \times \overline{\text{pre–sickness wage}} \times \text{disability grade},$$

where pre-sickness wage is capped at the minimum wage. Both benefits make flat-rate payments since they disregard how much individuals work once they are below or above the 50% threshold. Both benefits are paid as long as the individual is disabled but expire when the individual becomes entitled to the state pension at the statutory retirement age.[3] The individual can become entitled to UI benefit during the second stage of the WGA, after exhausting their UI benefit during the first-stage. The condition is to have worked 26 weeks in the last year and 4 years in the last 5 years.

These benefit rules imply two financial incentives to increase work effort when the first-stage wage-related benefit expires. The first incentive is due to the reduction in the amount of the DI benefit in the second-stage of the scheme since the UI benefit component of the first-stage wage-related benefit expires and is no longer a component of the second-stage wage-supplement or the follow-up benefits. The second incentive is due to the reduction in the DI benefit from the wage-supplement benefit to the follow-up benefit if individuals do not utilize at least 50% of their remaining work capacity in the second stage of the scheme: the wage-supplement benefit is based on pre-sickness wage while the follow-up benefit is based on $\overline{\text{pre–sickness wage}}$. As both incentives reduce the benefit level, they imply a strong negative income effect on labor supply. The size of the benefit reduction in the second-stage of the scheme, however, depends on how much individuals work in the first- and second stages, which implies a substitution effect.

[1] During the first two months of disability, the fraction is 0.75. It is possible for the utilization rate to be larger than 1 if remaining earning capacity is incorrectly estimated. In this case, the second component of the benefit is negative. If this situation persists, it may lead to a reassessment of the disability grade.

[2] The maximum duration has been reduced from 60 to 38 in January 2008, and it has been gradually reduced from 38 to 24 months in 14 quarters between January 2016 and April 2019.

[3] Individuals receive decision letters on DI eligibility from the Employee Insurance Agency on two occasions. First, they receive a letter on their application for the wage-related benefit at the end of their participation in the SI scheme. Second, they receive a letter approximately two months before their wage-related benefit expires which informs on the entitlement decision of a second-stage benefit.

Figure 1 illustrates the two financial incentives for a hypothetical individual who earns €100 per day before reporting sick and has a disability grade of 50%. It shows how the wage, DI benefit, and the sum of the two change with the amount of work. Amount of work is stated in terms of the daily wage during disability as a fraction of the pre-sickness daily wage. During disability, a daily wage equal to 25% of the pre-sickness daily wage implies a remaining work capacity utilization rate of 50% (given the assumed 50% disability grade). The figure distinguishes, with the vertical reference line, between two cases. To the left of the line, the individual utilizes less than 50% of her remaining work capacity in the second stage of the DI scheme and qualifies for the follow-up benefit. To the right of the line, she utilizes more than 50% of her remaining work capacity and qualifies for the wage-supplement benefit. The figure compares the total income from wage income (dashed or solid line in blue color) and a second-stage benefit (dashed or solid line in orange color) to the total income from earnings and the first-stage wage-related benefit (dashed or solid line in red color) at given remaining work capacity utilization rates.

Two notable patterns in Figure 1 demonstrate the financial incentives. First, a second-stage benefit is always smaller than the first-stage benefit. This demonstrates the first financial incentive: UI benefit is not part of a second-stage benefit. The difference (that is, the incentive) is smaller the more the individual works, and it vanishes if the individual utilizes her remaining work capacity to the full extent in which case the UI component of the first-stage benefit is 0 (where red line intersects orange line).

Second, a second-stage DI benefit is smaller if the individual utilizes less than 50% of her remaining work capacity (and qualifies for the follow-up benefit) than if she utilizes at least 50% of her remaining work capacity (and qualifies for the wage-supplement benefit). This demonstrates the second financial incentive: working less than the 50% threshold remaining work capacity utilization rate reduces the benefit. The vertical distance between the total income from earnings and the first-stage benefit (dashed or solid black line) and the total income from earnings and a second-stage benefit (dashed or solid gray line) shows the extent of the financial incentives to work in the second stage (when the individual utilizes less than or at least 50% of her remaining work capacity).

Individuals who earn higher pre-sickness wages face larger incentives to work because their benefit decreases by a larger amount from the first to the second stage of the DI scheme, implying larger income effects. This is illustrated by comparing Figure 1 to Figure A1 in the appendix. The financial incentives for individuals with high pre-sickness wages are not only larger than the incentives for those with low pre-sickness wages in absolute terms, but also relative to pre-sickness wages. Although the loss of the UI component of the wage-related benefit is proportional to the pre-sickness wage, the penalty for not meeting the threshold remaining earning capacity utilization rate of 50% is based on the difference between the pre-sickness wage and pre-sickness wage capped by the minimum wage ($\overline{\text{pre–sickness wage}}$). For pre-sickness wages below the minimum wage, this difference is non-existent. Above that, the difference increases with the pre-sickness wage.

The magnitude of the financial incentives may depend on whether the individual is entitled to the "social minimum supplement". This benefit supplements a social security benefit up to the subsistence-level "social minimum" set by the government.[4] However, the total of the social security benefit and the supplement cannot exceed the pre-sickness wage. If this prevents the individual from reaching the social minimum, she may apply for "social assistance" to supplement the income up to the social minimum. For some individuals the follow-up benefit can

[4]If an individual has a partner with an income, she may receive reduced or no social minimum supplement or social assistance. In addition, household assets such as savings are considered; exceeding the capital limit disqualifies one from receiving these benefits. Therefore, not everyone with a low income qualifies.

be very low so that they may qualify for the social minimum supplement and social assistance, which can compensate part of the reduction in the DI benefit in the second stage of the DI scheme. This situation is more likely to affect low pre-sickness wage earners, reducing their smaller incentives to resume work even more.

The financial incentives of the Dutch DI scheme are unique. In Norway and the US, for example, DI schemes allow earnings up to a given threshold, and decrease benefits if earnings exceed the threshold, at a rate of 60% in Norway and 50% in the US. In the first-stage of the Dutch DI scheme, there is no earnings allowance. That is, in case of earnings, the disability benefit is decreased at a rate of 70% (red line in Figure 1). This means that, although additional earnings lead to a higher total income from wages and the disability benefit (black line in Figure 1), the disincentive to work in the Netherlands is larger than those in Norway and the US. In the second stage of the Dutch DI scheme, earnings are allowed up to the full earnings capacity of the beneficiary, and as soon as earnings are higher than 50% of the remaining earning capacity, the disability benefit is increased (comparing solid and dashed orange lines in Figure 1). Moreover, the benefit is reduced in the second stage of the scheme since UI is no more a component of the disability benefit as in the first stage of the scheme (comparing red and orange lines in Figure 1). These are strong financial incentives to resume work that are unique to the Dutch DI scheme.

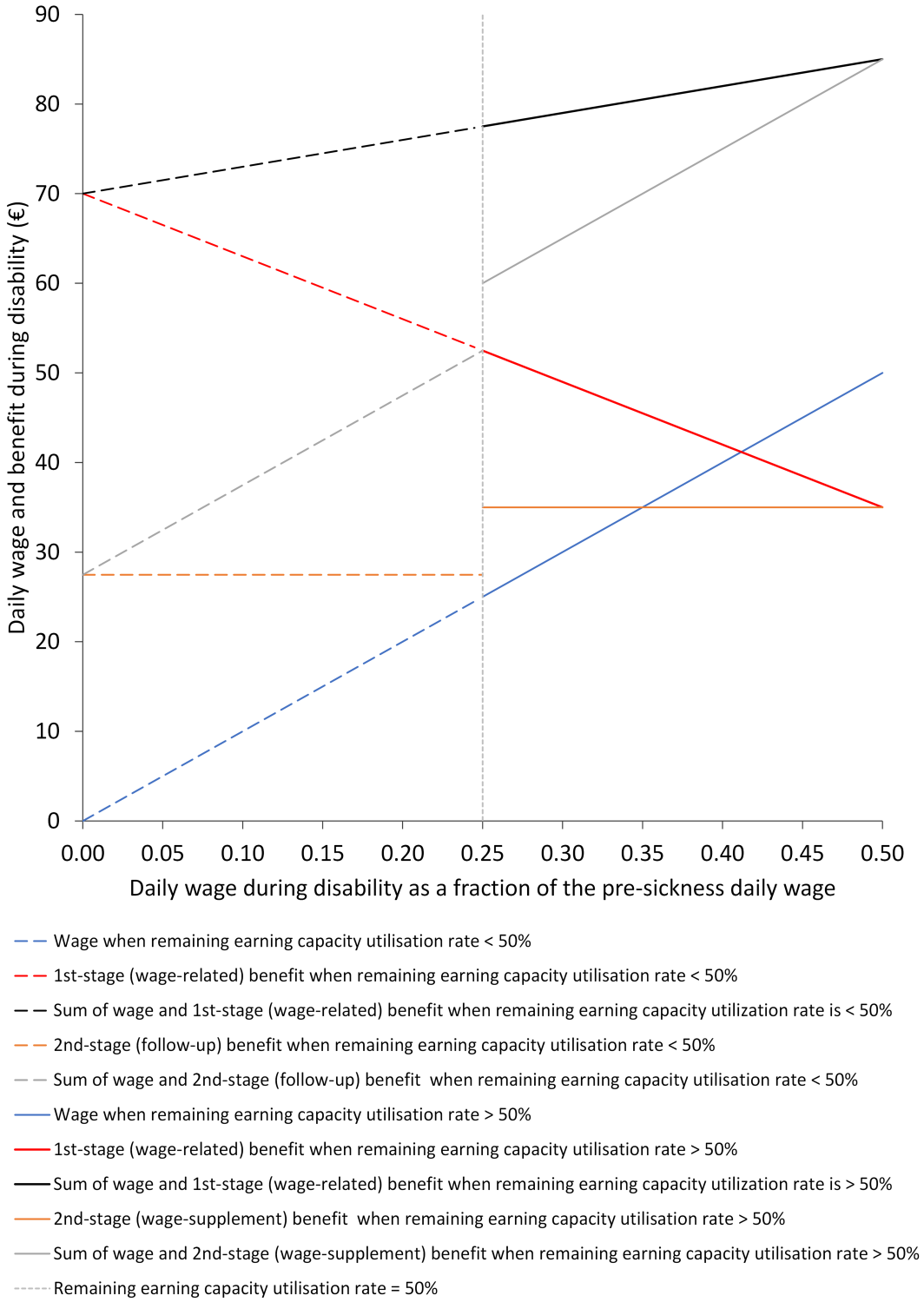


Figure 1: Financial incentives to work in the second stage of the DI scheme for an individual who earns €100 per day before falling sick and has a disability grade of 50%. The financial incentive is the difference between the dashed black line (sum of wage and the wage-related benefit) and dashed gray line (sum of wage and the follow-up benefit) if remaining work capacity utilization rate $< 50\%$ in the second stage of the DI scheme (left side of vertical reference line). The financial incentive is the difference between the solid black line (sum of wage and the wage-related benefit) and the solid gray line (sum of wage and the wage-supplement benefit) if remaining work capacity utilization rate $\geq 50\%$ in the second stage of the DI scheme (right side of vertical reference line).

3 Data

We use administrative data from the Employee Insurance Agency on all individuals who participated in the Return to Work Scheme for Partially Disabled People (WGA). The observation period begins in January 2006, when sick individuals could first apply for DI benefits under the new DI scheme (WIA), and ends in December 2020. We observe individuals' date of birth and gender, as well as, on a monthly basis, the type of DI benefit received (wage-related, wage-supplement, or follow-up), disability grade, and wages earned both before reporting sick and during DI receipt. At the time of reporting sick, we also observe whether individuals received unemployment insurance and, for wage earners, whether they held a permanent contract, a temporary contract, or a contract through a temporary work agency.

We link these individuals to administrative records from Statistics Netherlands, which provide monthly information on wages and benefits for the entire observation period. The benefits include the social minimum supplement, social assistance, and unemployment insurance.[5] Using the wage and benefit data, we track labor market participation and benefit receipt if individuals exit the DI scheme and thus leave the DI data.[6] We also link individuals to administrative data on their partners (married or cohabiting).

The initial data set contains 17,416,778 observations for 307,356 individuals who participated in the WGA. To select the estimation sample, we exclude individuals deemed temporarily fully disabled (Section 2).[7] These individuals do not face the financial incentives designed to encourage work resumption. We also exclude individuals who re-enter the WGA after exiting, and those who transition back to the first stage from the second stage. The resulting study sample comprises 3,504,283 observations for 57,437 individuals.

Based on the available data, we define five outcome variables: a dummy for labor market participation, based on earning positive wages; a dummy for receipt of the social minimum supplement; a dummy for receipt of social assistance; a dummy for receipt of unemployment insurance; and a dummy for the spouse's labor market participation.

4 Descriptive statistics on labor outcomes and benefit receipt

Table 1 reports sample averages of background and DI characteristics for individuals who remain in the sample until the financial incentive becomes relevant and are therefore observed at treatment. The average age is about 50 years, about half of the sample are men, and about half report having a spouse or partner. Permanent contracts are the most common employment type, with unemployment and temporary contracts each representing roughly half that share. Around 40% of the sample has a disability grade above 50%.[8] Individuals receive either the first-stage or second-stage benefit for almost two years on average, implying a total DI duration

[5] We do not observe receipt of the social minimum supplement after individuals leave the DI scheme because this information is only available within the DI administrative records.

[6] In our sample, 5.5% and 9.8% of individuals exit the DI scheme within 12 and 24 months, respectively, after the month in which the financial incentives take effect.

[7] These individuals have zero remaining earning capacity, implying minimal prospects of resuming work and limited responsiveness to financial incentives. More importantly, financial incentives are structurally absent: with zero remaining earning capacity, they automatically meet the threshold of earning at least 50% of that capacity, qualifying them for the higher wage-supplement benefit rather than the lower follow-up benefit in the second stage of the DI scheme. Additionally, because they do not receive unemployment insurance in the first stage, they are not subject to benefit reductions when it expires at the end of the first stage.

[8] Disability grade is defined as the difference between the reference wage and assessed remaining earning capacity, divided by the reference wage. The reference wage is the theoretical wage of someone who is not disabled but is similar to the partially disabled individual in all characteristics. The remaining earning capacity is based on a set of jobs that the partially disabled person can still do.

of around four years.[9] At disability onset, the average pre-sickness daily wage is about 118 euros (expressed in January 2006 prices).

The top panel of Figure 2 shows the fraction of individuals working during two periods of two years each. Months –24 to –1 refer to the period individuals claim the wage-related benefit. Month 0 indicates the first ("cut-off") month individuals claim the wage-supplement or the follow-up benefit and face the financial incentives to resume work.

The apparent decline in labor participation over event time reflects compositional changes in the population receiving first-stage DI benefits rather than declining employment among beneficiaries. Individuals with shorter entitlement durations have lower participation rates and are disproportionately represented at earlier event times. Consequently, aggregate participation appears to fall even though labor participation within duration groups increases over the course of the first-stage benefit period.

The middle and bottom panels of Figure 2 show the fractions of individuals receiving three types of benefits, as well as the fraction whose spouse is employed. There are clear and large jumps in the receipt of social minimum supplements and social assistance at the cut-off month. When distinguishing by pre-sickness wage group, even the highest wage group increasingly relies on these benefits once the DI benefit becomes less generous. This suggests that, regardless of prior earnings, some individuals struggle to resume work after exhausting their wage-related benefit and instead turn to alternative social support programs rather than earnings.

It is important to note that enrollment in these benefits is not automatic: individuals must file an application to claim them. Benefit substitution therefore cannot occur mechanically when DI benefits decrease at the cut-off month. Moreover, receipt of these benefits shows no anticipation effect. This pattern is consistent with the institutional rules: individuals must first exhaust their first-stage benefits before they become eligible for, and can apply for, these alternative benefits. Together, these patterns provide clear evidence that financial incentives induce behavioral responses.

It is not possible to claim UI while receiving the wage-related benefit (Section 2). However, a small fraction of the sample claims UI during the wage-related benefit period. These individuals may already have been receiving UI when they became partially disabled. UI claims increase gradually during the wage-supplement or follow-up benefit period. This pattern is consistent with individuals resuming work after exhausting their wage-related benefit (and hence their UI entitlement), but subsequently losing their job and becoming eligible for UI again.

The rightmost bottom panel shows the employment rate of spouses of DI recipients. Before the cut-off, spouse employment rises gradually, peaks at about 64% roughly one year before the cut-off, and then declines in the months leading up to it. After the cut-off, it fluctuates around 62 to 63%, with a slight downward trend. The confidence intervals overlap substantially over time, indicating that these changes are small. Overall, the figure provides no evidence that spouses adjust their labour supply in response to the change in DI incentives.

The event-time profiles in Figure 2 are based on unconditional averages and therefore do not account, for example, for individual fixed differences or common calendar-time effects. As a result, the profiles combine variation across event time with variation across individuals and calendar periods. In the next section we employ an event-study difference-in-differences framework that controls for these sources of variation.

[9] The duration of the first-stage wage-related benefit has varied over time due to reforms described in Section 2. The average of about two-year duration therefore does not imply that the sample has spent about the current statutory maximum number of months.

Table 1: Sample averages of background and disability characteristics

	Mean
Age	49.655
Male	0.530
With a spouse	0.552
Permanent contract	0.526
Temporary contract	0.221
Unemployed	0.253
Disability grade larger than 50%	0.398
Months spent in wage-related benefit	22.826
Months spent in wage-supplement benefit	20.544
Months spent in follow-up benefit	22.874
Pre-sickness daily wage	117.648
Number of individuals	46,245

Notes: 1. Except for the months spent receiving a DI benefit, all variables are measured in the last month of the first-stage benefit period. 2. Pre-sickness daily wage is expressed in January 2006 euros, adjusted for both price indexation and sector-specific wage inflation.

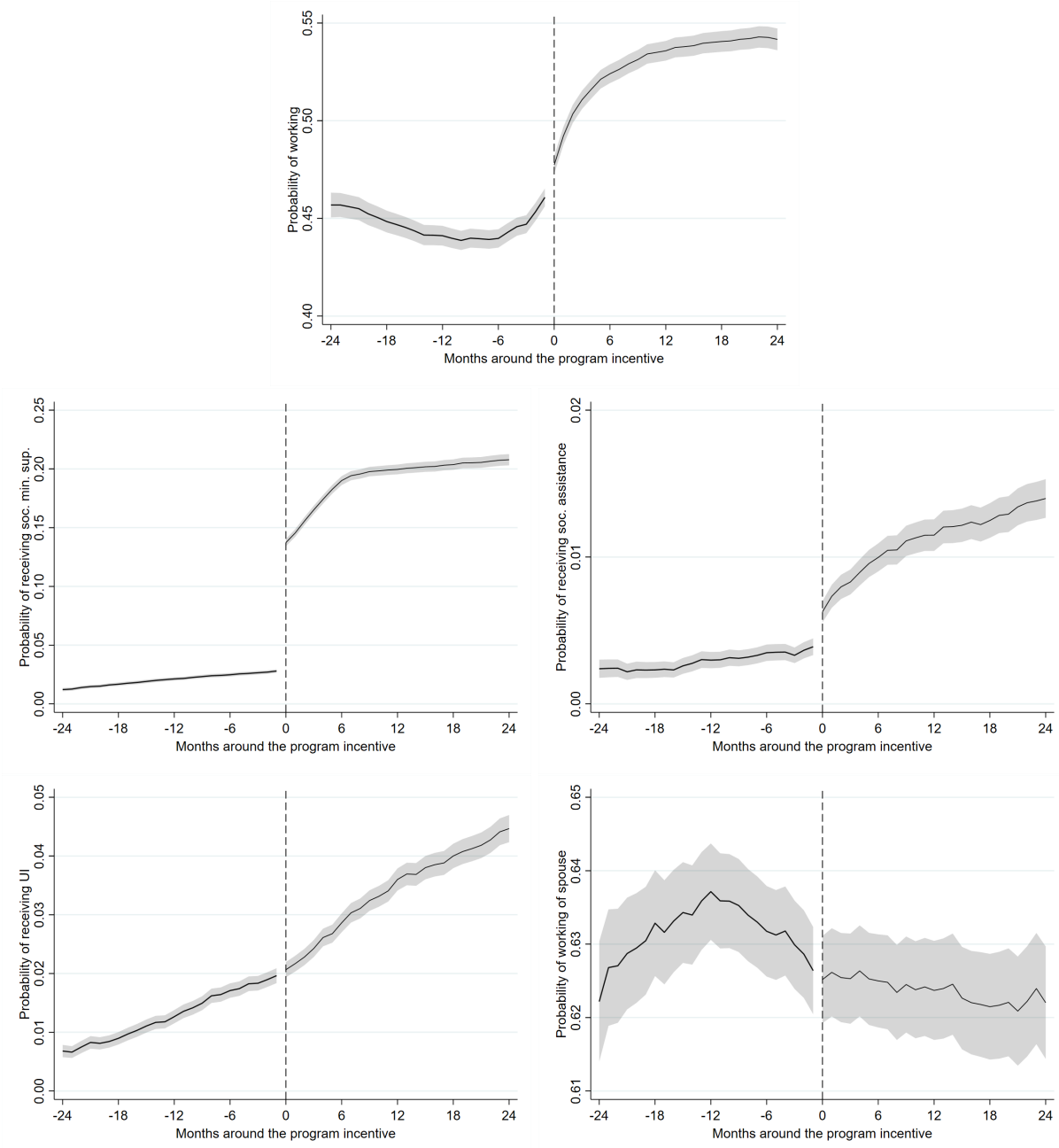


Figure 2: Labor participation, benefit receipt (social minimum supplement, social assistance, and unemployment insurance), and spousal labor participation during the wage-related benefit period and the subsequent period when the financial incentives become effective.

5 Identification strategy

To estimate the causal effect of financial incentives, we exploit variation in the timing of individuals' transition from the first-stage to the second-stage disability benefit. As described in Section 2, transition timing is determined by prior contribution histories. Figure 3 illustrates the resulting variation in first-stage benefit duration and, consequently, in exposure to the financial incentives embedded in the second-stage benefit. Because transition timing is predetermined by contribution histories and cannot be manipulated by beneficiaries, it provides a plausibly exogenous source of variation in exposure to financial incentives. We exploit this variation to identify the causal effects of financial incentives on labor supply and employment outcomes.

In settings with staggered treatment timing and potentially heterogeneous or dynamic treatment effects, standard Two-Way Fixed Effects linear regressions can yield severely biased estimates due to bad comparisons, namely, using early-treated units as controls for late-treated units (De Chaisemartin and D'Haultfœuille, 2020). To address this issue, we adopt the non-parametric difference-in-differences estimator for multiple time periods developed by De Chaisemartin and D'Haultfœuille (2020, 2026).

Let i denote the individual and t denote calendar quarters. The treatment variable, D_{it}, is a binary indicator taking the value of 1 starting two quarters prior to the expiration of the first-stage benefit, accounting for potential anticipatory behavior, and 0 otherwise. Identification compares switching cohorts at relative event time ℓ against a control group composed of not-yet-treated individuals (and never-treated individuals, if applicable), thereby avoiding the use of already-treated units as controls.

We estimate dynamic treatment effects δ_ℓ across relative time distance ℓ from the onset of financial incentives using the following dynamic specification:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell=-7}^{7} \delta_\ell \cdot \mathbb{I}\left(t - E_i = \ell\right) + \varepsilon_{it}, \tag{1}$$

where Y_{it} denotes the outcome of interest for individual i in quarter t. Outcomes include labor participation, benefit receipt from a social support program, and spousal labor market outcomes. E_i denotes the event quarter in which individual i first faces the financial incentives (D_{it} switches from 0 to 1). α_i is an individual-specific constant. It captures time-invariant labor participation or benefit claiming differences across individuals. It also captures differences in pre-sickness wage levels or time-invariant health conditions. γ_t absorb calendar-quarter fixed effects. ε_{it} is an idiosyncratic (unobserved) shock, assumed to be uncorrelated with all explanatory variables. Standard errors are clustered at the individual level to allow for arbitrary serial correlation and heteroskedasticity within individuals over time.

Our baseline specification evaluates four pre-treatment placebo leads ($\ell \in \{-7, \ldots, -4\}$). The period $\ell = -3$ serves as the reference period. The coefficients for $\ell \in \{-2, -1\}$ capture anticipation effects, while the coefficients for $\ell \in \{0, \ldots, 7\}$ capture post-treatment effects. In the figures, we only plot estimates for $\ell \in \{-7, \ldots, 5\}$ because estimates at longer horizons become increasingly imprecise as the number of available observations declines.

To prevent potential bias arising from non-random composition changes among switching cohorts over time, we restrict the estimation sample to a constant group of switchers observed across all post-treatment horizons, ensuring that dynamic effects at each relative horizon ℓ are estimated using a consistent panel of switching units across post-treatment periods.

Under the assumption of parallel trends conditional on individual and time fixed effects, the coefficients δ_ℓ for $\ell \geq 0$ identify the causal dynamic impact of financial incentives on labor participation and reliance on alternative social support programs.

The interpretation of placebo estimates differs across outcomes. For UI, in particular, pre-treatment coefficients are difficult to interpret because unemployment entitlements are incorporated into the wage-related disability benefit (Section 2). Consequently, UI receipt during the wage-related benefit period is largely governed by institutional rules rather than the labor-market behavior that is the focus of our analysis. After the transition to the second-stage benefit, however, individuals may again become entitled to UI if they return to work and subsequently satisfy the eligibility conditions for a new UI claim. We therefore place greater weight on the post-treatment dynamics when interpreting the effects on UI receipt.

As an additional validity check, we examine whether health status changes discontinuously at the benefit cut-off. Using administrative records on prescribed medicines as a proxy for health, we find little evidence of discontinuities around the transition from the first- to the second-stage benefit. These results suggest that the observed changes in labor supply and benefit receipt are unlikely to be driven by coincident changes in health status. Details are reported in the appendix.

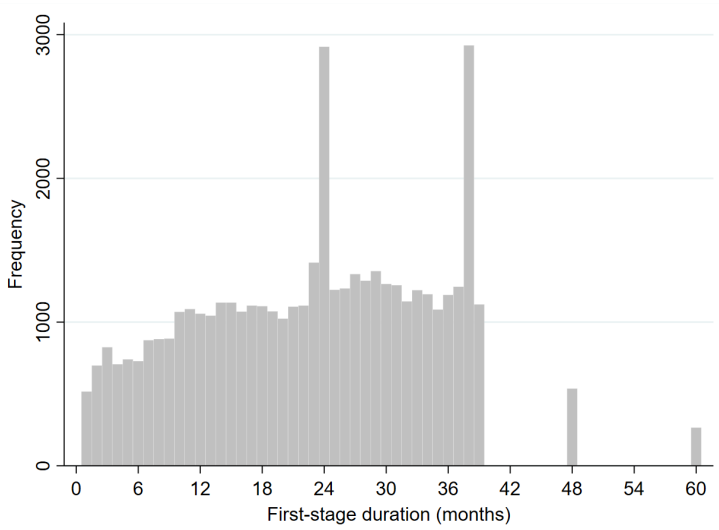


Figure 3: Frequency distribution of individuals by the number of months spent in the first stage of the disability insurance (DI) scheme before financial incentives take effect.

6 Effects of the financial incentives

6.1 Baseline effects

The top panel of Figure 4 presents the dynamic treatment effects on labor participation. The pre-treatment estimates are statistically indistinguishable from zero, supporting the parallel trends assumption. We document anticipatory responses of approximately 1 pp in the quarter immediately preceding the financial incentives. Following the onset of the financial incentives, labor-force participation increases immediately by about 3 pp and continues to rise gradually over the subsequent quarters, reaching a sustained effect of approximately 5 pp over the 18-month post-treatment period. The gradual increase in effects suggests an adaptation process. Overall, these findings demonstrate that financial incentives effectively motivate individuals to resume work.

As the financial incentives reduce DI generosity, some individuals may turn to alternative social support programs. The middle panel of Figure 4 provides evidence of this cross-program spillover: receipts of both the social minimum supplement and social assistance increase substantially and remain high across all post-treatment quarters. This finding aligns with existing literature demonstrating that DI reforms restricting program eligibility induce work resumption while simultaneously triggering participation in alternative social support programs to offset lost benefits (Karlström et al., 2008; Staubli, 2011; Borghans et al., 2014; Kantarcı et al., 2023).

We find small effects on UI receipt. This is expected because UI entitlement is generally exhausted during the wage-related benefit period (LGU). As discussed in Section 4, any observed increase in UI receipt is therefore likely driven by individuals who returned to work while receiving wage-supplement or follow-up benefits, subsequently lost their jobs, and qualified for a new period of UI entitlement.

Beyond individual labor supply and cross-program substitution, households may also rely on informal self-insurance through increased spousal labor participation. We find strong evidence of such a spousal response. This suggests that, while disabled individuals exhibit strong behavioral adjustments in both labor participation and benefit claiming, these responses do not fully offset the household income shock, maintaining the need for compensatory spousal labor supply. This finding aligns with Bernasconi et al. (2024), who document similar spousal responses following the 2006 DI reform in the Netherlands, and provides support for the added worker effect (AWE). Earlier studies examining the AWE in the context of spousal unemployment find limited evidence, often arguing that temporary job loss is too short-lived to trigger a labor supply response. The sizable effect we find suggests that spouses react when the income loss is persistent, which in our context is driven by health limitations.

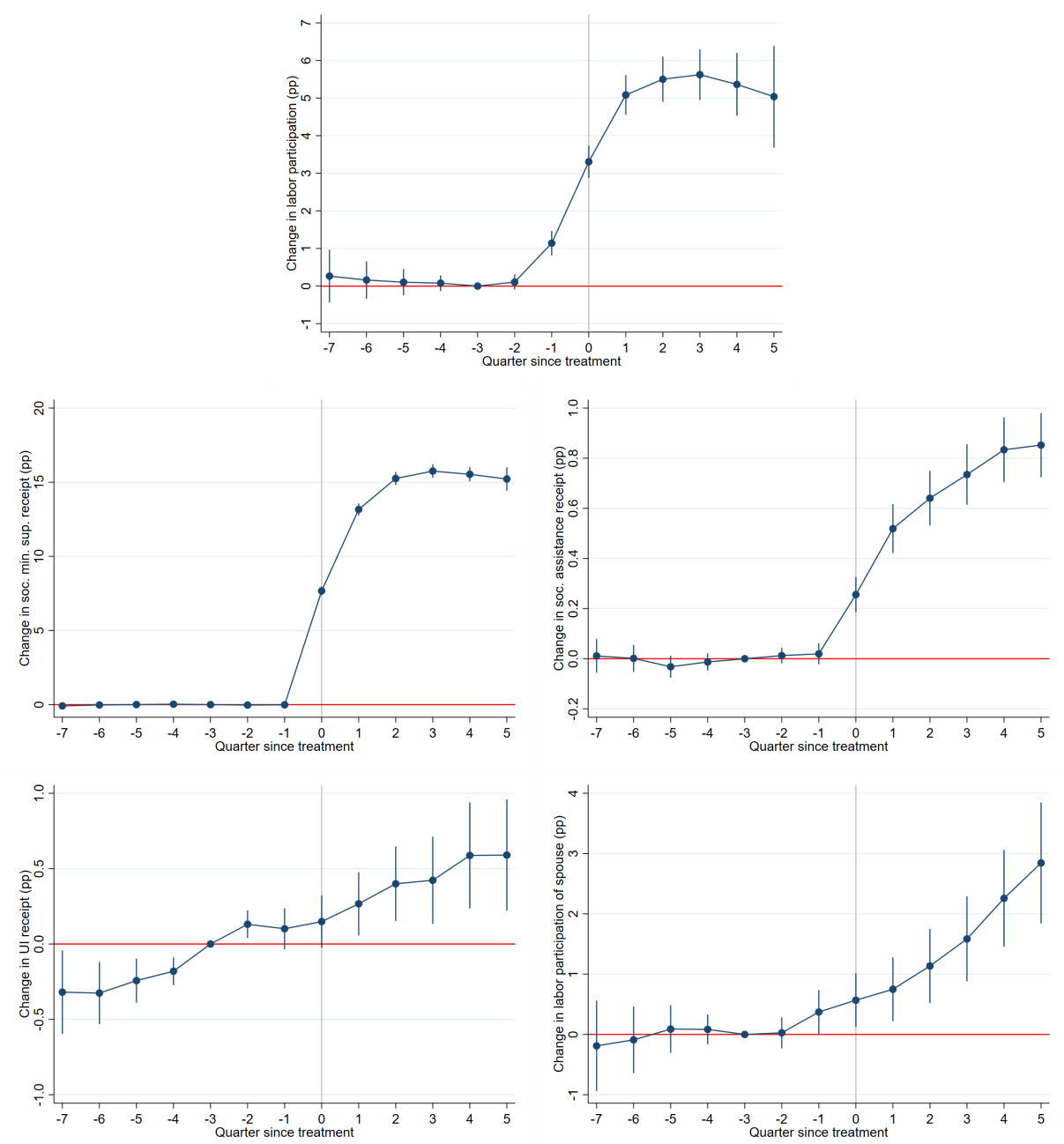


Figure 4: Estimated effects of the financial incentives on labor participation, benefit receipt, and spousal labor participation with 95% confidence intervals. Placebo joint nullity test p-values are 0.925 for labor participation, 0.790 for social minimum supplement, 0.386 for social assistance, 0.001 for UI, and 0.870 for spousal labor participation.

6.2 Do larger incentives induce larger responses?

As explained in Section 2, individuals with higher pre-sickness earnings face stronger financial incentives to resume work once their first-stage DI benefits expire. To exploit this source of institutional variation and assess whether responses differ across earnings levels, we stratify the sample into three pre-sickness wage groups in our baseline analysis.

Direct comparisons across earnings groups may, however, yield biased estimates of heterogeneous effects due to compositional differences in both observable and unobservable characteristics. To mitigate this concern, we identify a set of covariates that likely drive compositional differences across earnings levels and reweight the samples to align their covariate distributions. Because financial incentives become relevant only at the end of the first stage of the WGA, balancing is performed using characteristics measured at that point rather than at entry into the program. Consequently, the heterogeneity analysis is restricted to the 46,245 individuals who remain in the sample until the incentive becomes relevant. We construct the balancing weights using entropy balancing (Hainmueller, 2012), reweighting individuals in the low- and high-wage groups to match the first moments of the covariate distribution in the middle-wage group. As covariates, we consider age, gender, and contract type. We then estimate equation (1) using the resulting entropy-balancing weights.

Figure 5 shows that the financial incentives exhibit a wage gradient for labor market participation. Labor participation increases in all three wage groups following the transition to the second stage of the WGA. The estimated effect reaches approximately 4 to 5 pp among individuals with pre-sickness wages below 150% of the minimum wage, around 5 pp among those with wages between 150% and 250% of the minimum wage, and about 7 to 8 pp among those with wages above 250% of the minimum wage. The high-wage group exhibits both a larger immediate increase at the incentive cut-off and a higher long-run response. Although the confidence intervals overlap considerably across groups, limiting the statistical evidence for a pronounced wage gradient, the point estimates are nevertheless consistent with the institutional design of the scheme. Higher-wage earners experience larger reductions in benefit income when the wage-related benefit expires and therefore face stronger financial incentives to resume work. The larger responses observed for this group are consistent with these stronger incentives.

The increase in social minimum supplement receipt does not exhibit a monotonic wage gradient. The effect is largest for individuals with intermediate pre-sickness wages. A likely explanation is that eligibility for the supplement depends on both income and assets. Individuals with low pre-sickness wages are more likely to have already been eligible for the supplement before the financial incentives become effective, leaving less scope for further increases in receipt. Individuals with high pre-sickness wages, however, are less likely to satisfy the eligibility criteria.

Social assistance receipt exhibits a negative wage gradient. Higher pre-sickness wage earners claim social assistance less frequently, which is likely due to the means-tested eligibility requirements of the program.

We also find heterogeneity in spousal labor-supply responses across wage groups. While we do not find evidence of a spousal response among households with low pre-sickness earnings, the response becomes progressively stronger at higher wage levels. This pattern is consistent with the institutional design of the WGA, under which higher-wage earners experience larger reductions in disability benefits when the wage-related benefit expires. As a result, the shock to household income is larger for these households, increasing the incentive for spouses to compensate through greater labor market participation.

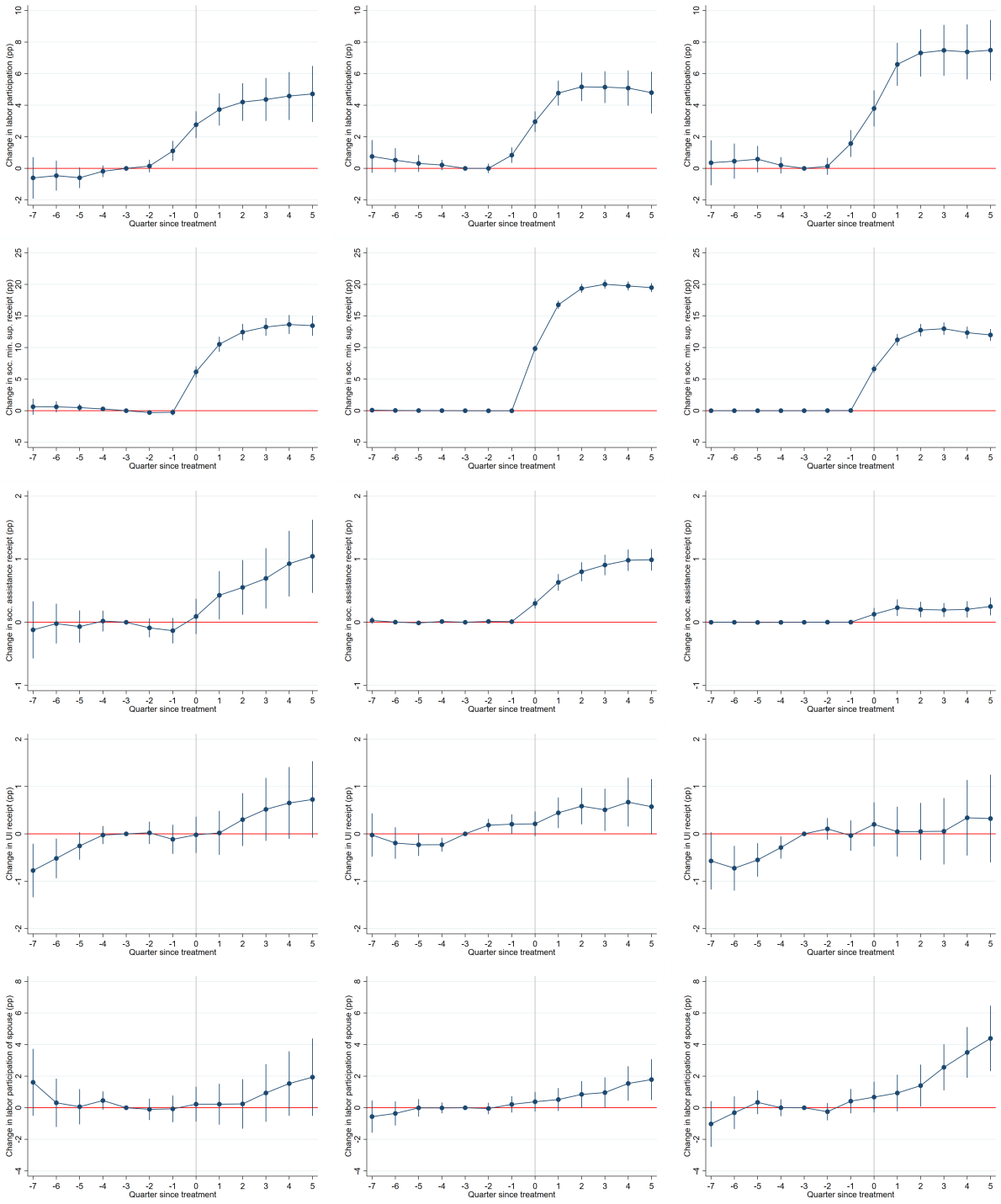


Figure 5: Estimated effects of financial incentives on labor participation, benefit receipt, and spousal labor supply with 95% confidence intervals across wage groups: less than 150% (left panel), between 150% and 250% (center panel), and more than 250% (right panel) of the minimum wage. Placebo joint nullity test p-values are 0.358, 0.568, and 0.744 for labor participation, 0.512, 0.627, and 1.000 for social minimum supplement, 0.735, 0.244, and 0.910 for social assistance, 0.091, 0.032, and 0.021 for UI, and 0.119, 0.746, and 0.246 for spousal labor participation, for individuals with pre-sickness wages below 150%, between 150% and 250%, and above 250% of the minimum wage, respectively.

6.3 Does employment status shape incentive responses?

Kantarcı et al. (2023) and Bernasconi et al. (2024) examined the effects of stricter eligibility criteria and reduced benefit generosity introduced by the Dutch DI reform of 2006. They find that the reform increased labor participation among individuals who became sick, but only if they held a permanent job. In contrast, spouses responded to the reform when the sick individual had a weaker labor market position, in terms of holding a temporary contract or being unemployed. These findings suggest that the impact of financial incentives may also vary across employment types, with potentially different responses among individuals with permanent contracts, temporary contracts, or no employment at the time of sickness onset.

As explained in Section 2, financial incentives to resume work for partially disabled individuals are strong, but they are offered only after a period of up to about 4 to 5 years after the first day of sickness. This delay complicates a successful return to the labor market. During the sickness period, individuals employed through temporary work agencies and those who are unemployed lack a former employer to return to, placing them at greater risk of unemployment compared to individuals with permanent contracts when they eventually face the financial incentives. In fact, in our data, three quarters prior to the incentive cut-off, the labor force non-participation rate is 40.3% among permanent contract workers, compared to 76.2% and 80% among temporary contract workers and the unemployed, respectively. If disabled individuals are not employed when they encounter the financial incentives, they may face barriers to re-entering the labor market, such as high job search costs, difficulty negotiating suitable work arrangements, or offered wages falling below their reservation wage (Koning and Lindeboom, 2015; Zaresani, 2020). For these reasons, financial incentives may be less effective among temporary contract workers and the unemployed. However, the notably lower labor participation rates observed in these groups prior to facing the incentives may also reflect substantial unused work capacity, suggesting that they could be more responsive to financial incentives.

Here, we examine whether financial incentives exhibit differential effects across employment statuses, comparing partially disabled individuals holding permanent contracts, those on temporary contracts, and those who are unemployed. To ensure comparability across these three groups, we reweight individuals in the temporary-contract and unemployed samples using entropy balancing so that the groups have identical means for age, gender, and pre-sickness wages.

Figure 6 shows substantial heterogeneity by contract type. Individuals without an employment relationship exhibit the strongest response to the expiration of first-stage DI benefits, with labor-force participation increasing by approximately 6-7 pp. Workers with permanent contracts also increase participation, although the effect stabilizes at around 3 pp. In contrast, workers with temporary contracts display only a transitory increase in participation, with the effect largely dissipating within a year after benefit expiration. As noted above, baseline participation rates are lower among individuals with temporary contracts and those without a contract, leaving greater scope for a behavioral response. While this may partly explain why these groups exhibit larger participation increases than permanent employees, it cannot explain the weaker response among temporary-contract workers relative to individuals without a contract.

Both the temporary-contract and no-contract groups claim alternative social support benefits more frequently. For temporary-contract workers, this may reflect the transitory nature of their labor-participation response, which largely dissipates within a year after benefit expiration despite an initial increase in participation. Among individuals without an employment relationship, the simultaneous increase in labor-force participation and reliance on alternative benefits may suggest substantial heterogeneity in adjustment paths, with some individuals successfully returning to work while others continue to depend on income support after DI benefits are reduced.

Spousal labor-supply responses are remarkably strong among partners of workers with permanent contracts but largely absent among the other groups. One potential explanation is that spouses of temporary-contract workers and individuals without an employment relationship already participate in the labor market at relatively high rates as a form of ex ante income insurance against their partner's precarious labor-market position, leaving less scope for additional labor-supply adjustment when financial incentives intensify following the expiration of first-stage DI benefits. However, the descriptive evidence does not support this explanation. Three quarters before benefit expiration, 65.2% of spouses in the permanent-contract group participated in the labor market, compared with 63.3% among spouses of temporary workers and 57.4% among spouses of individuals without an employment relationship. Instead, households with more precarious labor-market positions may have adopted other strategies to insure against income risk prior to disability onset, reducing the need for additional spousal labor-supply adjustments when first-stage DI benefits expire.

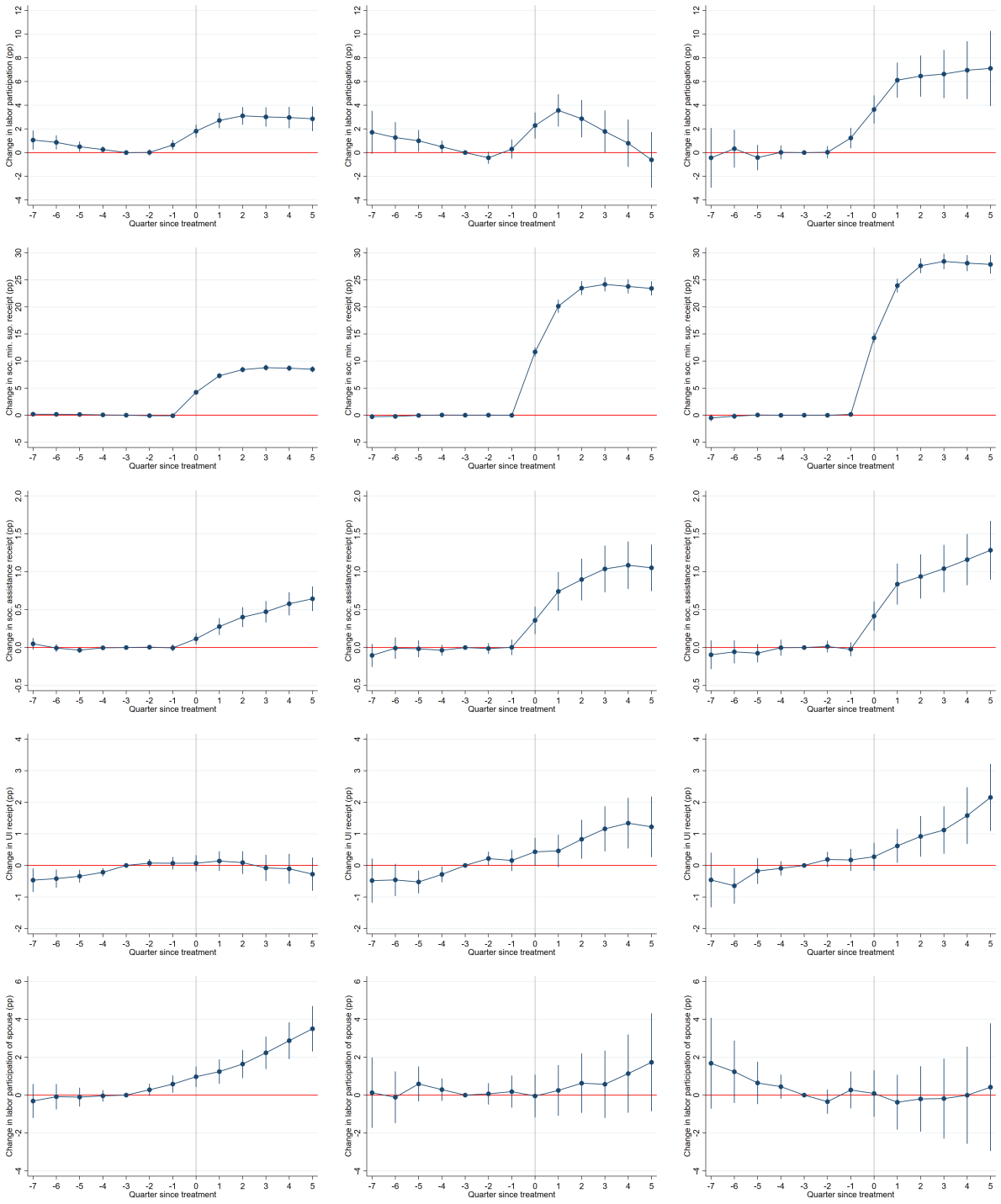


Figure 6: Estimated effects of the financial incentives on labor participation and benefit receipt with 95% confidence intervals for individuals who have a permanent contract (left panel), temporary contract (center panel), and who are unemployed (right panel). Placebo joint nullity test p-values are 0.055, 0.216, and 0.300 for labor participation, 0.271, 0.676, and 0.528 for social minimum supplement, 0.220, 0.221, and 0.645 for social assistance, 0.008, 0.062, and 0.139 for UI, and 0.901, 0.309, and 0.471 for spousal labor participation, for individuals who have a permanent contract, temporary contract, and who are unemployed, respectively.

6.4 Does labor demand affect incentive responses?

An adverse macroeconomic environment may reduce the effectiveness of financial incentives, as employment opportunities for individuals with disabilities are already limited and may deteriorate further during economic downturns. To capture labor market conditions, we use the vacancy rate as our primary measure of labor demand. Compared with aggregate indicators such as GDP growth, the vacancy rate provides a more direct measure of labor market tightness and the immediate availability of job opportunities. Figure 7 shows substantial variation in labor market tightness over the 2006-2020 sample period, with vacancy rates falling sharply during the Great Recession and increasing throughout the subsequent recovery. This variation generates meaningful differences in labor market conditions across individuals and allows us to examine whether the responsiveness to financial incentives depends on labor demand.

We define the vacancy rate as the end-of-quarter number of open job vacancies per 1,000 paid employee jobs.[10] Each calendar quarter is associated with a corresponding vacancy rate. For every individual, we calculate the average vacancy rate over quarters -2 to 5 by taking the mean of the vacancy rates observed during this period. Individuals are then classified according to whether this average lies above or below the median. To improve comparability, individuals in the above-median group are reweighted using entropy balancing such that both groups have identical means with respect to age, gender, and pre-sickness wages.

Figure 8 shows that work resumption responds substantially more strongly to financial incentives when labor demand is high than when it is low. The estimated effect is approximately three times larger in periods with above-median vacancy rates. Consistent with this pattern, claims for the social minimum supplement increase more strongly when labor demand is weak. Since the null of no pre-treatment effects is rejected for labor-force participation in the below-median vacancy-rate group, in the Appendix we present a sensitivity analysis to assess the robustness of these estimates to potential violations of the parallel-trends assumption.

The importance of labor demand is further illustrated by re-estimating Equation (1) using the 2006-2013 sample period studied by Koning and van Sonsbeek (2017). Figure 9 shows that the resulting treatment effects are substantially smaller than those obtained in our full sample. Given the weak labor demand during this period, the evidence suggests that labor market conditions help explain an important part of why Koning and van Sonsbeek found more modest responses to financial incentives.

Spousal labor-supply responses are larger when labor demand is high, suggesting that spouses are better able to offset income losses from lower disability benefits when employment opportunities are more abundant. However, the difference between high- and low-demand periods is not statistically significant.

[10] A vacancy is defined as any externally advertised position for which recruitment is actively underway and that is intended to be filled immediately or in the near future. This includes positions in the interview stage, temporary jobs, and paid apprenticeships, but excludes unpaid internships. Total jobs refer to all distinct paid employee positions.

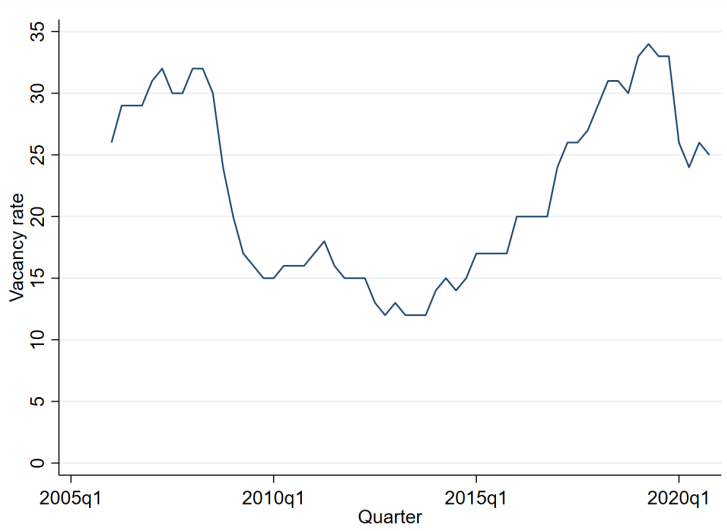


Figure 7: Quarterly vacancy rate, 2006q1-2020q4.

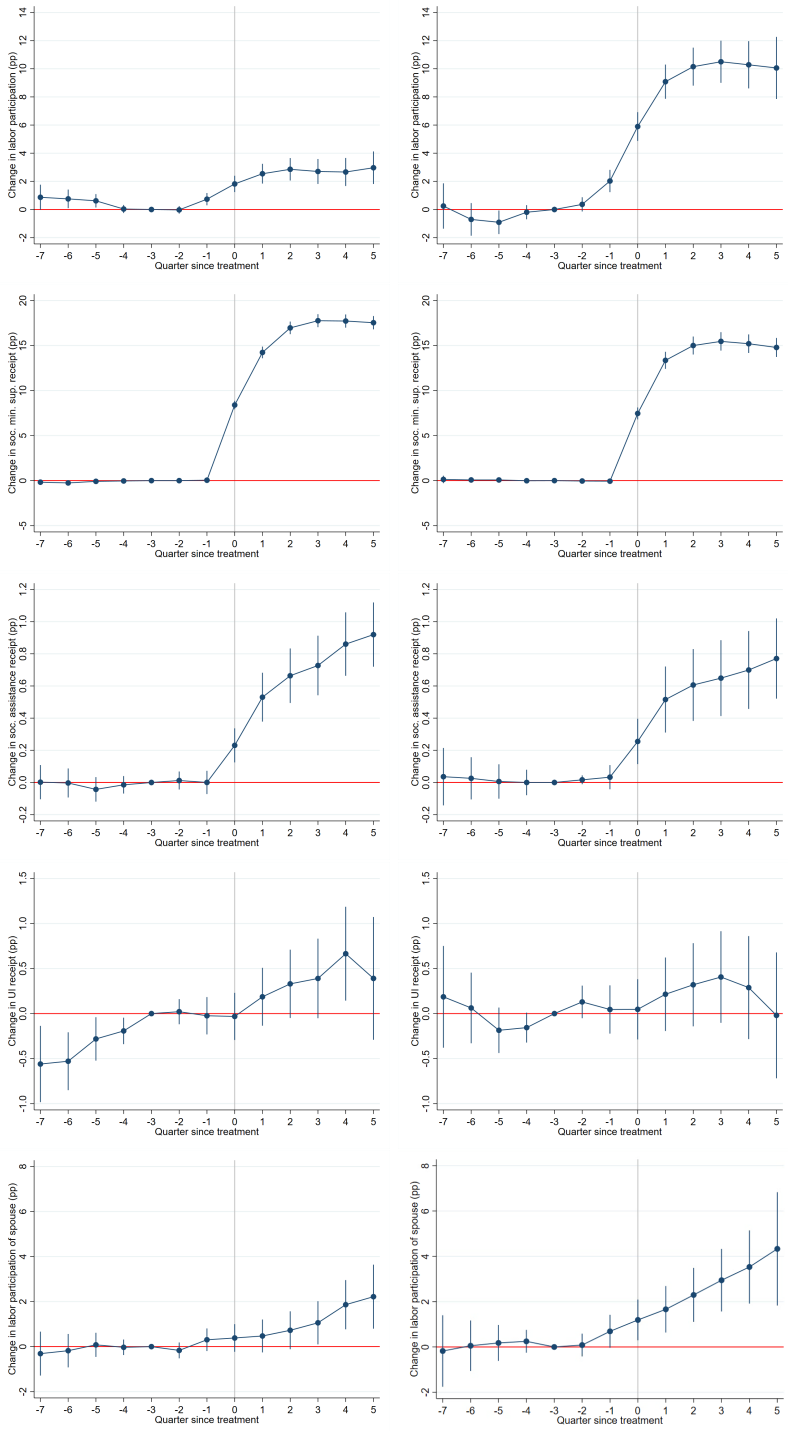


Figure 8: Estimated effects of financial incentives on labor participation, benefit receipt, and spousal labor supply with 95% confidence intervals across contemporary vacancy rate: below (left panel) and above (right panel) the median rate. Placebo joint nullity test p-values are 0.039 and 0.113 for labor participation, 0.324 and 0.924 for social minimum supplement, 0.620 and 0.996 for social assistance, 0.011 and 0.139 for UI, and 0.838 and 0.879 for spousal labor participation, for individuals exposed to financial incentives during periods with vacancy rates below and above the median, respectively.

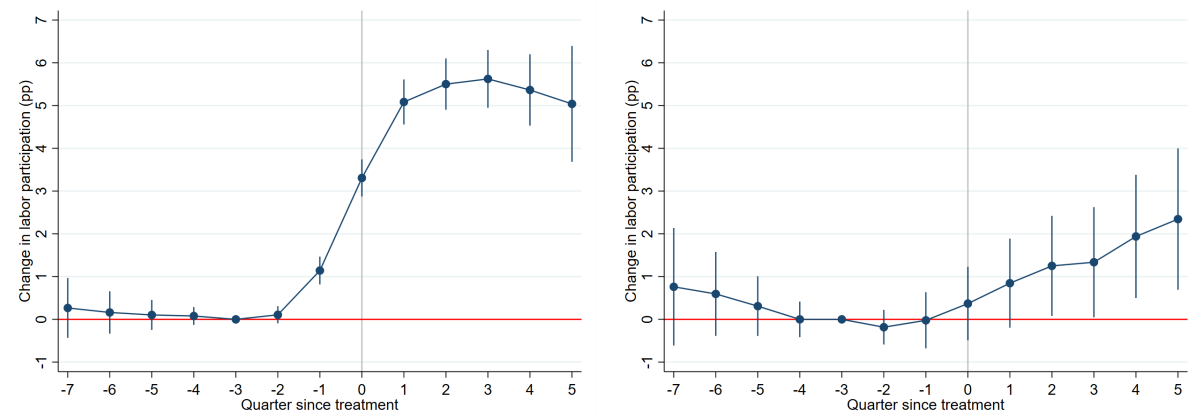


Figure 9: Estimated effects of financial incentives on labor participation. The left panel reproduces the baseline event-study estimates for the full 2006–2020 observation period. The right panel restricts the sample to the 2006–2013 period analyzed by Koning and van Sonsbeek (2017), corresponding to a period of relatively low labor demand.

7 The elasticity of labor force non-participation

The baseline results suggest that partially disabled individuals retain substantial work capacity and that financial incentives are effective in inducing them to utilize it. As described in Section 2, disability benefits are reduced discontinuously when the wage-related benefit expires and beneficiaries transition from the first to the second stage of the DI system. Moreover, the magnitude of this reduction depends on pre-sickness earnings and is institutionally determined. We exploit these design features to estimate the elasticity of labor-force non-participation with respect to benefit generosity. Specifically, we calculate the elasticity as the ratio of the observed change in labor-force non-participation between the first and second stages of the DI scheme to the percentage change in potential income over the same period. The latter captures the income loss that would occur if an individual did not adjust labor supply following the transition to the second-stage benefit. Expressing both changes in percentage terms yields an elasticity measure.

We calculate the observed change in labor-force non-participation as the negative of the treatment coefficient estimated for quarter 3 in our baseline specification. To express this change in percentage terms, we divide the estimated effect by the average non-participation rate in quarter 3. Because the panel is unbalanced, this average is computed by first averaging within individuals and then across individuals.

To calculate the percentage change in income, we compare income in the first and second stages of the DI scheme. First-stage income is defined as average total income in quarter -3, consisting of labor earnings, if any, and the wage-related DI benefit. The latter depends on pre-sickness earnings, the disability grade, and the utilization rate of remaining earning capacity. Pre-sickness earnings are observed, while the utilization rate is calculated as actual earnings divided by remaining earning capacity, where remaining earning capacity equals pre-sickness earnings multiplied by one minus the disability grade.

Second-stage income is defined as the income an individual would receive in quarter 3 if they did not work and relied exclusively on the follow-up benefit. In constructing this counterfactual, we assume that individuals do not anticipate changes in their disability grade and therefore hold the disability grade fixed at its first-stage value. The percentage change in income is then calculated as the difference between first-stage and second-stage income divided by average first-stage income.

We find that the elasticity of labor-force non-participation with respect to benefit generosity equals 0.136. This implies that a 10 percent reduction in benefit generosity reduces labor-force non-participation by approximately 1.36 percent. Existing estimates in the disability-insurance literature generally range between 0.1 and 0.3. Our estimate is close to those reported by Ruh and Staubli (2019) and Zaresani (2020), who estimate elasticities of 0.10 and 0.11 for Austria and Canada, respectively. It is somewhat smaller than the estimates reported by Marie and Vall Castello (2012) and Deuchert and Eugster (2019), who find elasticities of 0.22 and 0.26, respectively, and lies within the range of 0.1 to 0.3 reported by Kostøl and Mogstad (2014) for Norwegian disability-insurance reforms. Overall, these comparisons suggest that labor-supply responses among partially disabled individuals in the Netherlands are broadly in line with those documented in other disability-insurance systems.

Table 2 presents elasticity estimates by pre-sickness wage group. Consistent with our baseline findings on labor-force participation, higher pre-sickness wage earners are more responsive to financial incentives. This is unsurprising given that the potential income loss associated with the transition to the second-stage benefit is substantially larger for these individuals. The last column of Table 2 illustrates the much greater magnitude of these incentives. Notably, higher-wage earners exhibit larger elasticities despite having lower baseline non-participation rates prior to the onset of the financial incentives, as shown in the third column of Table 2. This

means that even among individuals who are already relatively likely to participate in the labor market, work-resumption decisions remain highly responsive to financial incentives.

Table 2: Elasticity of labor force non-participation by pre-sickness wage group

Pre-sickness wage as a fraction of the minimum wage	Elasticity of labor non-participation	Mean labor non-participation during quarter -3 (%)	Potential daily incentive (€)
Smaller than 150%	0.115	61.0	41.9
150-250%	0.121	56.8	83.8
Larger than 250%	0.178	50.9	137.1

8 Conclusion

Targeting beneficiaries in a DI scheme is crucial for policy interventions, as measures affecting the existing stock of beneficiaries can yield a significantly greater immediate impact than those aimed at the relatively small pool of applicants (Borghans et al., 2014). Because partially disabled individuals retain a portion of their work capacity, DI programs for this population provide a natural setting for policies designed to encourage labor market participation (De Jong and de Vos, 2005). Partially disabled individuals may, however, engage in moral hazard by claiming benefits for work capacity that they do not utilize. Although the Employee Insurance Agency examines the remaining work capacity of all sick-listed workers at the end of the SI period, changes in work capacity are rarely reexamined during the subsequent DI period. This is because the Agency can conduct only a limited number of reexaminations due to shortages of medical examiners and the substantial costs associated with reexamining large numbers of beneficiaries. Moreover, it is difficult to observe whether partially disabled individuals actively search for work but fail to receive job offers, making it challenging to verify continued benefit eligibility. Financial incentives may therefore be an effective instrument for promoting work resumption because they do not require extensive reexamination procedures.

We analyzed the effects of financial incentives built into the Dutch DI scheme for partially disabled individuals. For the average (pre-sickness) wage earner in our population, these financial incentives increase labor force participation by about 5 pp. This finding suggests that financial incentives are effective at promoting work resumption and that partially disabled individuals retain substantial work capacity. In fact, the effects are larger among individuals who were unemployed than among workers with permanent contracts (at the time of reporting sick), despite the former generally being considered to have fewer employment opportunities and to be particularly vulnerable in the labor market. These findings suggest that moral hazard may be an important determinant of labor supply among partially disabled individuals and that financial incentives can mitigate it. This contrasts with Silva and Vall-Castelló (2017), who show that, despite the possibility of combining disability benefits with work, disabled individuals in Spain exhibit substantially lower job-search intensity than their nondisabled counterparts. Their model attributes a 32 pp gap in search intensity to lower productivity and suggests that hiring subsidies are unlikely to effectively activate unused work capacity.

Financial incentives are less effective at inducing work resumption among the more than one-quarter of DI recipients who are low-wage earners. By design, this group faces relatively weak financial incentives to return to work. Moreover, low-wage earners have poorer health status (Kantarcı et al., 2026), while limited skills may further constrain their employment opportunities, reducing their ability to respond to these incentives. However, the financial consequences of the incentive scheme remain substantial even for high-wage earners. They respond more strongly to the incentives because the incentives are larger for this group. In addition, they rely on alternative social support programs, and even their spouses increase labor supply to partly offset the resulting income losses. These findings suggest that whether financial incentives ultimately increase inequality remains an open question.

We find that financial incentives are substantially more effective when labor demand is high. This result suggests that disabled individuals face important demand-side constraints and that financial incentives alone may be insufficient to stimulate work resumption during periods of weak labor demand. Consequently, the effectiveness of activation policies depends not only on the incentives they provide but also on prevailing labor market conditions.

We find no evidence that stronger financial incentives worsen health outcomes, alleviating concerns that work resumption may be achieved at the expense of beneficiaries' health.

We find strong substitution between disability insurance (DI) benefits and alternative social

support programs. When exposed to stronger financial incentives, partially disabled individuals increase labor market participation but also rely more on alternative benefit programs to offset income losses. While alternative benefits partly cushion these income losses, they generally do not eliminate spousal labor-supply responses. Spouses complement formal income support when income losses are larger and benefit durations are longer, particularly when favorable macroeconomic conditions strengthen labor demand.

The business cycle plays a critical role in shaping the effectiveness of financial incentives, highlighting the vulnerable position of disabled individuals in the labor force. During recessions and periods of labor market slack, their employment prospects diminish markedly. Conversely, the stronger employment responses observed during economic expansions suggest that favorable labor market conditions may reduce the extent to which disabled individuals are disadvantaged in hiring and retention. These findings highlight the importance of incorporating macroeconomic conditions into the design and evaluation of reintegration policies.

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Appendix

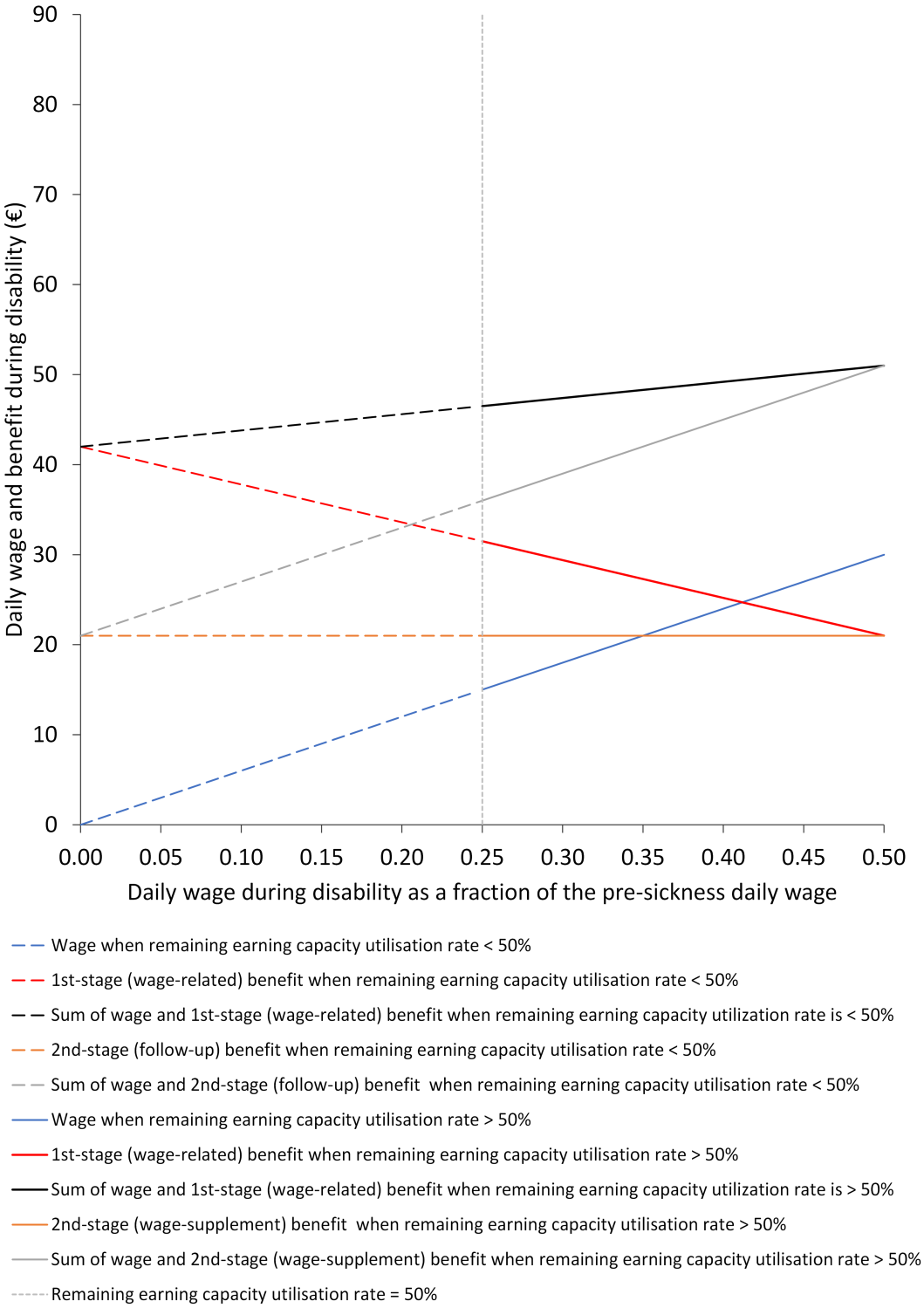


Figure A1: Financial incentives to work in the second stage of the DI scheme for an individual who earns €60 per day before falling sick and has a disability grade of 50%. Dashed lines apply when remaining work capacity utilization rate $< 50\%$. Solid lines apply when remaining work capacity utilization rate $\geq 50\%$. The financial incentive is the difference between the dashed black line (sum of wage and the wage-related benefit) and dashed gray line (sum of wage and the follow-up benefit) if remaining work capacity utilization rate $< 50\%$ in the second stage of the DI scheme (left side of vertical reference line). The financial incentive is the difference between the solid black line (sum of wage and the wage-related benefit) and the solid gray line (sum of wage and the wage-supplement benefit) if remaining work capacity utilization rate $\geq 50\%$ in the second stage of the DI scheme (right side of vertical reference line).

Validity check using prescribed medicines

The identifying assumption underlying our event-study design is that, in absence of financial incentives, treated and not-yet-treated individuals would have followed parallel trends in the outcomes of interest. While this assumption is not directly testable, its plausibility can be assessed by examining whether predetermined characteristics evolve systematically around the onset of treatment. We focus on health as a key potential confounder and investigate whether health status exhibits changes around the transition from the first- to the second-stage DI benefit.

We focus on health as a key covariate. In the Netherlands, the mandatory health care insurance (Zorgverzekeringswet) reimburses costs of prescribed medicines. Data on prescriptions are available from Statistics Netherlands. Using this information, we compute the annual number of prescribed medicines, treating higher counts as indicative of poorer health. We then estimate equation (1) using this health proxy as the outcome variable. A statistically significant treatment effect would indicate that health status changes systematically around the transition from the first- to the second-stage benefit, raising the possibility that changes in health, rather than financial incentives, contribute to the estimated labor-market responses.

A caveat is that information on prescribed medicines is only available annually, whereas the treatment indicator varies at the monthly level. As a result, it is not possible to align health outcomes precisely with the month of transition to the second-stage DI benefit. We therefore aggregate the analysis to the annual level and define event time relative to the calendar year in which individuals first face the financial incentives. This ensures that each individual contributes to a unique event-time period in the dynamic treatment-effect analysis. The calendar year of first exposure is defined as year 0, regardless of whether the transition occurs early or late in the year, and the preceding calendar year serves as the base year.

Figure A2 presents the estimated treatment effects. The number of prescribed medicines does not exhibit a discontinuous change at or after the cut-off. The estimated effects are not only statistically indistinguishable from zero but also economically negligible, with point estimates of approximately -0.02 prescribed medicines.

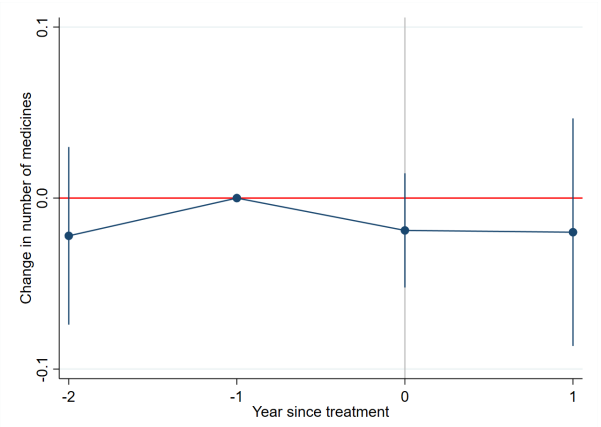


Figure A2: Estimated effects of financial incentives on number of medicines prescribed with 95% confidence intervals.

Honest DiD

We conduct a sensitivity analysis following Rambachan and Roth (2023) using the event-study estimates of De Chaisemartin and D'Haultfœuille (2020). The pre-treatment period is defined as quarters -7 to -3, while quarters -2 and -1 are treated as anticipation periods and excluded from the analysis. The parameter of interest is the average treatment effect across post-treatment quarters 2 to 5. We impose the smooth-differences restriction, which allows for post-treatment deviations from parallel trends but requires that the evolution of these deviations remains smooth relative to the pre-treatment period. Figure A3 reports the original confidence intervals alongside identified sets obtained under progressively larger departures from parallel trends, corresponding to sensitivity parameter values of M equal to 0, 0.01, 0.02, 0.03, 0.04, 0.05 and 1. The identified sets for the above-median and below-median groups begin to overlap only for values of M equal to 0.04 or larger. Under our calibration, 0.04 roughly corresponds to allowing the slope of the parallel-trends violation to approximately double by the post-treatment horizon. We view substantially larger values of M as difficult to reconcile with the relatively smooth pre-treatment dynamics.

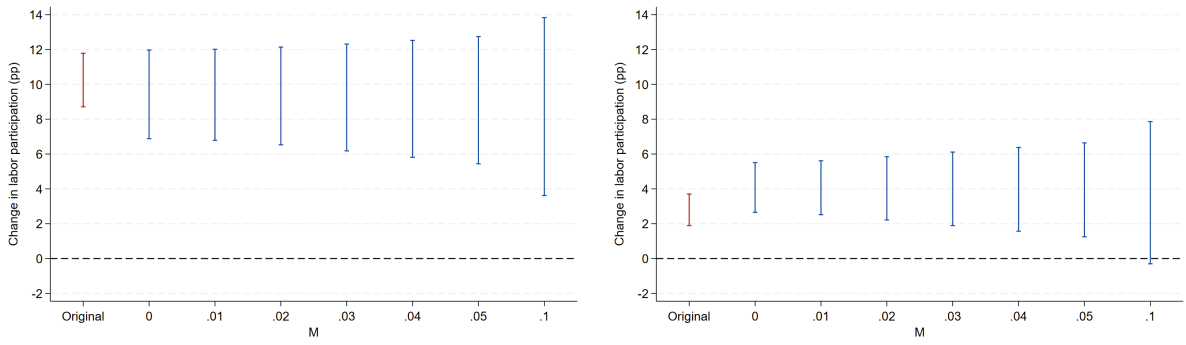


Figure A3: Treatment-effect bounds under alternative smooth-differences restrictions.